Binh Minh Tran

Honours Student, Computational Data Science, School of Computer Science, The University of Sydney

Contact Information

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Personal Profile

I am interested in a simple but difficult question: how can we build machine-learning models that remain useful when real patients, biological measurements, and clinical settings are heterogeneous?

I am a final-year Bachelor of Advanced Computing (Honours) student at the University of Sydney, working at the intersection of machine learning, biomedical data, and computational medicine. My research has led me from graph learning and multimodal prediction to spatial single-cell imaging, genomics, and survival modelling, with a growing focus on how heterogeneous patient information can be integrated to support more reliable and individualised prediction.

What motivates me most is not simply improving predictive performance, but understanding how models behave when data are incomplete, populations differ, or measurements come from different biological and clinical settings. I am particularly interested in multimodal learning, graph-based representations, uncertainty, and cross-cohort generalisation.

My long-term goal is to develop robust and interpretable machine-learning methods for precision medicine that can bridge methodological innovation with meaningful biomedical and clinical questions.

Education

Feb 2023 – Dec 2026

Bachelor of Advanced Computing (Honours), Computational Data Science
The University of Sydney, Sydney, Australia.
Thesis: *"Hierarchical Niche-Aware Multimodal Learning for Breast Cancer Survival Prediction Using Imaging Mass Cytometry, Clinicopathologic Variables, and Gene-Level Data"*, supervised by Associate Professor Xiuying Wang, School of Computer Science.
Cumulative WAM 82/100 (Distinction); First Class Honours expected, mark and class released December 2026.

2019 – 2022

High School Diploma, Specialised Physics Programme
HUS High School for Gifted Students (HSGS), VNU University of Science, Vietnam National University, Hanoi, Vietnam.
GPA 9.5/10; outstanding academic achievement award, academic years 2019–2022.

Research Interests

Biomedical AI **Graph Learning** **Multimodal Learning** **Spatial Omics**
Computational Oncology **Machine Learning** **Deep Learning**

Publications

† Equal contribution. Applicant's name in bold. Quartiles: SCImago (SJR) 2025; ranks: CORE.

Journal Publications

[J1] **B. M. Tran**†, V. A. Nguyen†, L. H. Nguyen, Q. A. Dang, N. M. H. Nguyen, Q. A. Nguyen, T. B. V. Nguyen, and D. H. Tran, "Discrimination was retained but calibration was not: development and external evaluation of a multimodal deep-learning model for breast cancer receptor classification," *BMC Medical Informatics and Decision Making* (SJR Q1), *under review*, 2026.

[J2] **B. M. Tran**, D. Q.-A. Nguyen, M.-H. Ha, D. M. Pham, T. D. Nguyen, Q. B. Doan, D. T. Doan, M.-A. Nguyen, D.-T. Nguyen, and O. T.-C. Chen, "H2F-GCN: Skeleton-based Hypergraph Frequency Coordinate Fusion Graph Convolution Network for Physical Rehabilitation Assessment," *IEEE Access* (SJR Q1), *under review*, 2026.

[J3] M.-H. Ha, X.-H. Le, A.-T. Dang, D.-P. Dao, H.-M.-D. Vu, D.-T. Kim, **B. M. Tran**, and O. T.-C. Chen, "GRAU++: An Efficient Lightweight Ghost-Residual-Attention U-Net++ Architecture for Automated Angle of Progression Biometry," *EAI Endorsed Transactions on AI and Robotics* (SJR Q1), *under review*, 2026.

[J4] **B. M. Tran**, V. A. Nguyen, Q. A. Dang, and N. M. H. Nguyen, "Development and External Validation of a Robust Deep Learning Model for Benign–Malignant Breast Lesion Classification on Ultrasound," *Journal of Medical Research* (Hanoi Medical University), *under review*, 2026.

Peer-Reviewed Conference Publications

[C1] **B. M. Tran**†, V. A. Nguyen†, L. H. Nguyen, Q. A. Dang, N. M. H. Nguyen, Q. A. Nguyen, H. N. Tran, T. B. V. Nguyen, and D. H. Tran, "A Multimodal Deep Learning Framework for Breast Cancer Receptor Status Prediction," in *Proc. 12th International Symposium on Integrated Uncertainty in Knowledge Modelling and Decision Making (IUKM 2026)*, Quy Nhon, Vietnam, Sep. 2026. Springer LNAI vol. 16918 (Part I), 13 pp. *In press.* doi:10.1007/978-981-92-5641-9_9 Full paper

[C2] L. H. Nguyen, **B. M. Tran**, and D. H. Tran, "A Variational Graph Neural Network with Multi-view Integration for Predicting lncRNA–Disease Associations," in *Recent Challenges in Intelligent Information and Database Systems (ACIIDS 2026)*, CORE B, Communications in Computer and Information Science, vol. 2964, Springer, pp. 355–369, 2026. *Published.* doi:10.1007/978-981-92-0068-9_24

[C3] L. H. Nguyen, **B. M. Tran**, and D. H. Tran, "Dual-Stream Contrastive Graph Representation Learning for lncRNA–Disease Association Prediction," in *Proc. FISU Joint Conference on Artificial Intelligence (FJCAI 2026)*, Can Tho, Vietnam, Mar. 2026, pp. 151–158. *Published.* ISBN 978-604-45-2586-0

[C4] **B. M. Tran**, L. Zhang, and X. Wang, "HiNT-Surv: Hierarchical Niche-Aware Multi-Microenvironment Learning from Imaging Mass Cytometry for Survival Prediction," in *IEEE Symposium on Computational Intelligence in Health and Medicine (CIHM), IEEE Symposium Series on Computational Intelligence (SSCI) 2027*, *submitted*.

Other Publications

[O1] **B. M. Tran**, "Promoting the Role of Overseas Vietnamese Youth in Protecting the Party's Ideological Foundation in the Digital Space" (in Vietnamese), *State Management Review*, Academy of Administration and Public Management, Hanoi, Vietnam, 17 Aug. 2026. *Published.* Full text

Research Experience

2026 – **Incoming AI Research Intern**, GeneStory (Vingroup), Hanoi, Vietnam — on-site.

- Vingroup's genomics and precision-medicine company, built on data from the Vietnamese 1,000 Genomes Project (VinBigData); gene-sequencing services for disease risk, medication response and nutrition.

Feb 2026 – Present

Honours Research Student, School of Computer Science, The University of Sydney, Sydney, Australia. Supervisor: A/Prof Xiuying Wang.

- Designed HiNT-Surv, a hierarchical niche-aware model fusing imaging mass cytometry with clinicopathologic and genomic evidence: availability-masked additive fusion decomposes each patient's risk exactly into modality and core–niche contributions, an antigen-identity encoder transfers across antibody panels, and all data-dependent steps stay inside each training fold to prevent patient-level leakage.
- On 404 METABRIC and 285 Basel patients (834 tissue cores), patient-level nested cross-validation gave the highest mean Harrell's C on METABRIC (0.740 $\pm$ 0.020 vs 0.718) and the lowest integrated Brier score on Basel (0.140 $\pm$ 0.013 vs 0.156), separating Kaplan–Meier risk groups in both cohorts (log-rank $p < 10^{-8}$); METABRIC-to-Basel transfer without retraining held the highest Harrell's C point estimate (0.700) while the closest multimodal comparator fell to 0.485.

Dec 2025 – Present

Research Collaborator, Department of Obstetrics & Gynaecology, Hanoi Medical University, Hanoi, Vietnam. Multimodal breast-cancer receptor prediction.

- Co-developed a multimodal, multitask deep-learning framework integrating clinical variables, H&E whole-slide images and tumour RNA-seq for joint ER, PR and HER2 prediction; reported in [C1].
- Evaluated with patient-level stratified five-fold cross-validation on 937 TCGA-BRCA cases (macro AUC 0.890; ER 0.954, PR 0.868, HER2 0.848) and frozen external testing on 122 CPTAC-BRCA cases without retraining, tuning or threshold adjustment (macro AUC 0.931).

Dec 2025 – Mar 2026

Research Assistant, Center for Biomedical and Pharmaceutical Research, Vietnam Academy of Science and Technology (VAST), Hanoi, Vietnam.

- Applied reproducible NGS preprocessing and annotation workflows (FastQC, Trimmomatic, genome assembly, gene prediction and functional annotation) to mutation and microbial gene-function analyses.

Research Projects

Dec 2026 – Nov 2027

Development and Calibration of an AI Model for Individualised Post-Test Probability after High-Risk NIPT (Trisomy 21, 18 and 13)

- Role: designing and calibrating the probability model — a retrospective cohort of high-risk NIPT referrals, converting a screen-positive label into an individualised probability of confirmed fetal aneuploidy from signed Z-score, fetal fraction and assay platform.
- Principal investigators: Nguyen Van An and **Binh Minh Tran**; supervisor: Nguyen Thi Bich Van, MD, PhD, Department of Obstetrics & Gynaecology, Hanoi Medical University. Institutional research scheme, 12 months, total budget 10,000,000 VND — *proposal submitted.*

Awards & Scholarships

2026 — **Finalist (Top 10)**, AURORA Scientific Research Awards 2026, Vietnam–Australia Innovation Network (NIC-AU) — shortlisted for the Grand Final, Hanoi, Dec. 2026.

2026 — **Outstanding Student Award**, Embassy of Vietnam in Australia — for academic and research achievement.

2026 — **Second Prize**, Vietnamese Young Intellectuals Forum in Australia — for the multimodal breast-cancer receptor-status research [C1].

2026	**Third Prize**, Youth Science and Technology Conference, Hanoi Medical University — for [C1]; work selected for the 23rd Youth Science and Technology Conference of the Vietnamese Health Sector.
2025	**January Star Award**, Vietnam National Union of Students — one of 117 national recipients, for academic achievement, research and student leadership.
2023 – 2026	**Sydney International Student Award**, The University of Sydney — merit-based tuition scholarship awarded at admission and held for the full degree.

Industry Experience

Dec 2024 – Mar 2025	**Data Engineer Intern**, Viettel Software, Hanoi, Vietnam. • Built and maintained Python/SQL ETL pipelines for operational datasets; contributed to data-quality monitoring, schema design and query/pipeline performance tuning supporting analytics teams.
Jun 2024 – Jul 2024	**Data Engineer Intern**, Agribank (Vietnam Bank for Agriculture and Rural Development), Hanoi, Vietnam. • Supported ETL of banking datasets into reporting systems; contributed to dashboards and operational reports used by internal stakeholders.

Technical Skills

Programming	Python, R, SQL, Bash.
Deep learning	PyTorch, PyTorch Geometric, scikit-learn.
Methods	Graph neural networks, self-supervised and contrastive learning, multimodal fusion, survival modelling.
Evaluation	Patient-level stratified cross-validation, bootstrap confidence intervals, paired hypothesis testing, multiple-testing correction, effect-size analysis, external validation.
Biomedical data	Spatial single-cell IMC, RNA-seq / gene expression, histopathology whole-slide images.
Research computing	Git, Linux, experiment tracking.
Languages	Vietnamese (native); English (fluent — all tertiary study in English).

References

Referee 1	**Associate Professor Xiuying Wang** School of Computer Science, The University of Sydney, Australia. Email: xiu.wang@sydney.edu.au
Referee 2	**Associate Professor Dang Hung Tran** Hanoi University of Industry, Hanoi, Vietnam. Email: hungtd@haui.edu.vn
Referee 3	**Dr Manh Hung Ha** Faculty of Applied Sciences, International School, Vietnam National University, Hanoi (VNU-IS). Email: hunghm@vnu.edu.vn